

LIBXS Samples

Batched GEMM

Exercises the LIBXS batched GEMM API (`libxs_gemm.h`) in several flavors: strided, pointer-array, indexed, and grouped. All programs are multithreaded via OpenMP, report wall-clock time and GFLOPS/s, and optionally validate results via `libxs_matdiff`.

Building

```
cd samples/gemm
make GNU=1
```

LIBXS must be built first from the repository root.

Programs

gemm_strided Strided batch DGEMM: all matrices packed contiguously with constant stride. Uses `libxs_gemm_index` with `index_stride=0`.

```
./gemm_strided.x [M [N [K [batchsize [nrepeat [beta [pad]]]]]]]
```

Defaults: M=N=K=23, batchsize=30000, nrepeat=3, beta=1.0, pad=0.

gemm_batch Pointer-array batch DGEMM: matrices accessed through pointer arrays. Supports a duplicate C-matrix mode to exercise lock-forward sync.

```
./gemm_batch.x [M [N [K [batchsize [nrepeat [dup [beta [pad]]]]]]]]]
```

Defaults: M=N=K=23, batchsize=30000, nrepeat=3, dup=0, beta=1.0, pad=0.

dup	Mode	Description
0	None	Each C-matrix is unique (<code>LIBXS_GEMM_FLAG_NOLOCK</code>)
1	Sorted	Half-unique C-pointers, duplicates consecutive
2	Shuffled	Half-unique C-pointers, randomly shuffled

gemm_index Index-array batch DGEMM: matrices in contiguous buffers, addressed through explicit element-offset index arrays (`ia`, `ib`, `ic`).

```
./gemm_index.x [M [N [K [batchsize [nrepeat [beta [pad]]]]]]]
```

Defaults: M=N=K=23, batchsize=30000, nrepeat=3, beta=1.0, pad=0.

gemm_indexf Fortran variant of `gemm_index`. Demonstrates the LIBXS Fortran module interface with one-based index arrays and `C_LOC/C_SIZEOF` interop. Requires a Fortran compiler.

```
./gemm_indexf.x [M [N [K [batchsize [nrepeat]]]]]
```

Defaults: M=N=K=23, batchsize=30000, nrepeat=3.

gemm_groups Grouped batch DGEMM: multiple groups of different matrix shapes dispatched in sequence, each with its own `libxs_gemm_config_t`. Per-group dimensions grow by 4 starting from `base_m`.

```
./gemm_groups.x [ngroups [batch_per_group [nrepeat [base_m [beta [pad]]]]]]]
```

Defaults: ngroups=2 (max 4), batch_per_group=30000, nrepeat=3, base_m=8, beta=1.0, pad=0.

Validation

Set the CHECK environment variable to validate results:

```
CHECK=1 ./gemm_strided.x
CHECK=1 ./gemm_batch.x 23 23 23 1000 1 2
CHECK=1 ./gemm_index.x
CHECK=1 ./gemm_groups.x
```

When padding is enabled (pad>0), the check also verifies that leading-dimension padding in C-matrices has not been overwritten.

Memory Comparison Benchmark

Benchmarks libxs_diff, libxs_memcmp, and the C standard library's memcmp for both large contiguous comparisons and many small fixed-size comparisons. A Fortran variant compares libxs_diff against Fortran array intrinsics.

Building

```
cd samples/memory
make GNU=1
```

Produces memcmp.x (C) and, if a Fortran compiler is found, memcmpf.x and matcpyf.x.

Usage (C)

```
./memcmp.x [elsize [stride [nelems [niters]]]]
```

Argument	Default	Description
elsize	INSIZE/MAXSIZE (compile, 64)	Element comparison size in bytes
stride	max(MAXSIZE, elsize)	Distance between successive elements
nelems	2 GB / stride	Number of elements
niters	5	Repetitions (arithmetic average)

Environment Variables

Variable	Default	Description
STRIDED	0	0: one-call when stride==elsize, else loop; 1: seq loop; >=2: random
CHECK	0	Non-zero: validation pass (inject diffs, cross-check implementations)
MISMATCH	0	0: equal buffers; 1: first byte; 2: middle; 3: last; >=4: random
OFFSET	0	Shift buffer b off its natural alignment by this many bytes

Compile-Time Knobs

Macro	Default	Description
MAXSIZE	64	Stride floor (elements spaced >= this far)
INSIZE	MAXSIZE	Default element size when argument is omitted

```
## 32-byte elements, sequential strided access, 10 repetitions
./memcmp.x 32 0 0 10

## 8-byte elements, random traversal, 3 repetitions
STRIDED=2 ./memcmp.x 8 0 0 3

## contiguous (single-call) comparison, ~2 GB
./memcmp.x 64 64 0 5

## mismatch at last byte, buffer b misaligned by 3 bytes
MISMATCH=3 OFFSET=3 ./memcmp.x 32 0 0 5
```

Environment variables and compile-time macros apply only to `memcmp.x`.

Usage (Fortran -- memcmpf)

```
./memcmpf.x [nelements [nrepeat]]
```

Element type is hard-coded as `INTEGER(4)`. Compares ~2 GB by default. Reports per-iteration times (ms) and average throughput (MB/s).

Usage (Fortran -- matcopyf)

```
./matcopyf.x [m [n [ldi [ldo [nrepeat [nmb]]]]]]
```

Argument	Default	Description
m	4096	First matrix dimension
n	m	Second matrix dimension
ldi	m (enforced m)	Leading dimension of input
ldo	ldi	Leading dimension of output
nrepeat	2	Number of repetitions
nmb	2048	Memory budget in MB

Benchmarks `libxs_matcopy` and `libxs_matcopy_task` for matrix copy and zeroing. When the matrix is square and `ldi==ldo`, also benchmarks `libxs_otrans` and `libxs_otrans_task` for out-of-place transpose.

What Is Measured

Three comparison implementations are benchmarked:

- `libxs_diff` -- SIMD-dispatched short-buffer comparison (SSE/AVX2/AVX-512). Limited to `elsize < 256` bytes, per-element.
- `libxs_memcmp` -- SIMD-dispatched `memcmp` replacement. Single call for contiguous buffers, or per-element for strided access.
- `stdlib memcmp` -- C library implementation under the same pattern.

Each kernel gets freshly initialized data before its timed section. An untimed warm-up pass runs before the first measured iteration. The summary reports min / median / max across iterations plus peak MB/s. Reported MB/s counts both buffers ($2 * \text{nbytes} / \text{time}$).

Ozaki-Scheme Low-Precision GEMM

Intercepts DGEMM, SGEMM, ZGEMM, and CGEMM at link time, replacing them with low-precision Ozaki schemes (mantissa slicing or CRT). Real GEMM calls go through the selected scheme; complex GEMM (ZGEMM/CGEMM) is mapped to real GEMM internally.

Build

```
cd samples/ozaki
make GNU=1 -j $(nproc)
```

LIBXS must be built first from the repository root. When a sibling `libxstream` directory is detected, the optional OpenCL/GPU path is compiled in (see `OZAKI_OCL` below).

Link-Time Interception

Two link-time variants are built per precision:

- `dgemm-blas.x` / `sgemm-blas.x` -- dynamically linked against BLAS
- `dgemm-wrap.x` / `sgemm-wrap.x` -- linked with `--wrap=` (GNU ld)
- `zgemm-wrap.x` / `cgemm-wrap.x` -- complex GEMM wrappers

Static (`libwrap.a`) and shared (`libwrap.so`) wrapper libraries are built by default. The shared library can intercept any application:

```
LD_PRELOAD=/path/to/libwrap.so ./application
```

The static library requires explicit `--wrap` flags:

```
-Wl,--wrap=dgemm_ -Wl,--wrap=sgemm_ -Wl,--wrap=zgemm_ -Wl,--wrap=cgemm_
```

Test scripts: `test-wrap.sh` exercises all variants, `test-check.sh` validates both schemes, `test-mhd.sh` tests MHD file-based input.

Test Driver

```
dgemm-wrap.x [A.mhd|M [B.mhd|N [K [TA [TB [ALPHA [BETA [LDA [LDB [LDC]]]]]]]]]]
```

Defaults: `M=N=K=257`, `TA=TB=0`, `ALPHA=BETA=1.0`. The first argument can be 0 followed by MHD filenames to load A and B matrices from files. The `zgemm-wrap.x` and `cgemm-wrap.x` drivers call ZGEMM and CGEMM.

Environment Variables

Scheme Selection

Variable	Default	Description
OZAKI	2	0=bypass (BLAS), 1=mantissa slicing, 2=CRT, 3=adaptive
OZAKI_COMPLEX	(auto)	Complex dispatch: 0=BLAS, 1=CPU, 2=GPU+fallback. Auto: 2 if on
OZAKI_N	(auto)	Slices (Sch.1: fp64=8, fp32=4) or primes (Sch.2: fp64=16, fp32=9)

OZAKI=3 (adaptive) starts with Scheme 1 on the first GPU call to learn the effective cutoff from preprocessing occupancy. Subsequent calls compare the Scheme-1 pair count (at the cached cutoff) against the Scheme-2 prime count and pick the cheaper path. On CPU, adaptive falls back to Scheme 2.

Accuracy

Variable	Default	Description
OZAKI_FLAGS	3	Sch.1 bitmask: 1=Triangular, 2=Symmetrize, 0=full S^2 square
OZAKI_TRIM	0	Levels to trim (0=exact). ~7 bits/level (Sch.1), ~4 bits (Sch.2)
OZAKI_I8	0	Sch.2: signed i8 residues (moduli<=128) instead of u8
OZAKI_GROUPS	0	Sch.2: K-grouping (0/1=off). Consecutive K panels, one reconstr.
OZAKI_MAXK	32768	Max K per preprocessing pass (0=full K in one pass)
OZAKI_THRESHOLD	12	Intensity threshold. Bypass when flops/(bytes*thr)<1. 0=always

GPU Path (requires LIBXSTREAM)

Variable	Default	Description
OZAKI_OCL	0	Enable OpenCL/GPU path (0=off, 1=on)
OZAKI_TM	(auto)	GPU output tile height (multiple of 8)
OZAKI_TN	(auto)	GPU output tile width (multiple of 16)

GPU-specific kernel tuning variables (OZAKI_RTM, OZAKI_RTN, OZAKI_WG, OZAKI_SG, OZAKI_KU, OZAKI_RC, OZAKI_PB, OZAKI_HIER, OZAKI_PREFETCH, OZAKI_SCALAR_ACC, OZAKI_DEVPOOL, OZAKI_CACHE) are documented in the LIBXSTREAM Ozaki README.

Monitoring and Diagnostics

Variable	Default	Description
OZAKI_VERBOSE	0	0=silent, 1=stats at exit, N=print every Nth GEMM call
OZAKI_STAT	0	Track C-matrix (0), A-representation (1), or B-representation (2)
OZAKI_DUMP	0	Dump A/B as MHD files at the given call-count (0=off)
OZAKI_EPS	inf	Dump A/B when epsilon error exceeds threshold
OZAKI_RSQ	0	Dump A/B when RSQ drops below threshold (updated after dump)
OZAKI_EXIT	1	Exit on accuracy violation after dump. 0=continue
OZAKI_PROFILE	0	Profile: 0=off, 1=all, 2=kernel, 3=preprocess A, 4=preprocess B

Benchmark

Variable	Default	Description
CHECK	0	Validate vs BLAS: 0=off, negative=auto-threshold, positive=custom
NREPEAT	3	Number of GEMM calls (first call is warmup when >1)
EVIL	0	Adversarial exponent-span test (see below)
OZAKI_DECAY	0	Decay diagnostic + K-permutation (0-4, see below)

Adversarial Input (EVIL) The EVIL variable initializes A and B with controlled exponent structure for stress-testing accuracy and adaptive slice reduction. The magnitude sets the exponent span in bits; the sign selects the distribution:

EVIL=N (N>0) Per-column. Column j of A is scaled by $2^{(N*j/(ncols-1))}$, column j of B by the inverse. Product A*B is well-conditioned. Uniform exponents within each column.

EVIL=-N (N>0) Per-element. Each element gets a pseudorandom exponent in [0,N] via coprime shuffle, with opposite sign for B. Every row of A spans the full exponent range -- worst case for row-wise alignment and adaptive cutoff.

EVIL=0 Default shuffle mode (no exponent structure).

The per-column mode (EVIL>0) matches the NVIDIA emulation grading test (diagonal scaling with D and D^-1). The per-element mode (EVIL<0) is adversarial for the adaptive slice-pair reduction: it forces all slices to be populated in every row.

Decay Diagnostic and K-Permutation (OZAKI_DECAY) Controls K-dimension row reordering for smoothness and the forward-difference decay diagnostic. Values map to libxs_sort_t:

Value	Behavior
0	Off (default). No permutation, no diagnostic.
1	Identity permutation. Measure decay on original ordering.
2	Sort B rows by L1-norm, apply same K-permutation to A.
3	Sort B rows by mean value, apply same K-permutation to A.
4	Greedy nearest-neighbor chain on B rows ($O(K^2*N)$).

Values 2-4 compute a K-permutation from B (via libxs_sort_smooth) before Ozaki-1 slicing. Both A and B are sliced using the permuted K-order. Since $C = A*B = (A*P^T)*(P*B)$ for any permutation matrix P, the result is mathematically identical -- only the int8 digit structure along K changes.

When nonzero, reports the forward-difference decay of int8 slice buffers along K, M, and N axes (first K-group only, single- threaded). Uses libxs_fprint per-axis mode internally. Output goes to stderr:

OZ1[MxNxK] Delta-K: d1=... d2=... ... OZ1[MxNxK] Delta-M: d1=... d2=... ... OZ1[MxNxK] Delta-N: d1=... d2=...
...

Decaying values indicate exploitable smoothness; growing values (~2x per order) indicate unstructured data where data-independent schemes (Ozaki-1, Ozaki-2) are well-matched.

Profiling

The OZAKI_PROFILE variable enables per-GEMM timing collected into a histogram reported at program exit:

OZAKI_PROF: 850 DP-GFLOPS/s (17.0 INT8-TOPS/s, 20x)

Profile modes select which phase is measured:

Mode	CPU	GPU
1/negative	All phases (preprocess+dot)	All profiled kernels
2	Kernel only (dot products)	Dotprod kernel only
3	Preprocessing (A+B)	Preprocess A kernel
4	Preprocessing (A+B)	Preprocess B kernel

On the CPU, modes 3 and 4 are equivalent (A and B preprocessing is interleaved across OpenMP threads).

Example

```
./dgemm-wrap.x 256 # CRT scheme (default)
OZAKI=1 ./dgemm-wrap.x 256 # mantissa slicing
OZAKI_TRIM=4 OZAKI=1 ./dgemm-wrap.x 256 # drop 4 least significant diagonals
OZAKI=2 OZAKI_GROUPS=4 ./dgemm-wrap.x 4096 # CRT with K-grouping
OZAKI=3 ./dgemm-wrap.x 4096 # adaptive scheme selection
EVIL=512 ./dgemm-wrap.x 1024 # accuracy grading (wide exponent span)
EVIL=1 OZAKI_PROFILE=1 ./dgemm-wrap.x 1024 # narrow span (shows pair savings)
EVIL=-52 ./dgemm-wrap.x 1024 # per-element span (worst for cutoff)
OZAKI_DECAY=1 OZAKI=1 ./dgemm-wrap.x 256 # decay diagnostic (no reorder)
OZAKI_DECAY=2 OZAKI=1 ./dgemm-wrap.x 256 # sort by norm + decay diagnostic
OZAKI_DECAY=4 OZAKI=1 ./dgemm-wrap.x 256 # greedy NN + decay diagnostic
```

Predict Samples

Four executables demonstrating fingerprint-guided prediction:

- **predict_params** -- Parameter prediction from structured CSV (GPU kernel tuning, configuration databases).
- **predict_sunspots** -- Timeseries forecasting via sliding-window kNN (monthly sunspot numbers, 1749-present).
- **predict_earthquakes** -- Spatial prediction of earthquake magnitude from location and depth (USGS catalog).
- **predict_discharge** -- River discharge forecasting via sliding-window kNN with day-of-year seasonality (USGS NWIS daily streamflow).

Build

```
make
```

Or from the LIBXS root:

```
make GNU=1 samples/predict
```

predict_params

Train a prediction model from a CSV file and save it for later use. Finds the optimal training fraction and polynomial order automatically. Reports validation quality on a held-out subset.

```
./predict_params.x [fraction] [auto|cat|interp] [-N] <csvfile> [modelfile]
```

fraction	Validation split 0..1 for quality report (default: 0.8). The full model always trains on all entries.
auto	Auto-detect mode per output (default).
cat	Force categorical (kNN) for all outputs.
interp	Force interpolation for all outputs.
-N	Max polynomial order for final build (default: 0 = auto).
csvfile	Delimited text file (semicolons, commas, or tabs). The first line may be a header (auto-skipped if non-numeric).
modelfile	Output path for the binary model. Default: derived from CSV basename (e.g., data.csv -> data.bin).

```
./predict_params.x ../../samples/smm/params/tune_multiply_PVC.csv
```

predict_sunspots

Timeseries forecasting using sliding-window nearest-neighbor prediction. The recent history (window of W values) serves as input; the next H values are predicted as output. The kNN confidence indicates whether similar patterns were seen in training.

```
./predict_sunspots.x <csvfile> [train_fraction]
```

```
csvfile          Semicolon-delimited timeseries (SILSO sunspot format).
train_fraction   Fraction of data used for training (default: 0.8).
```

```
./predict_sunspots.x predict_sunspots.csv 0.8
```

```
Loaded 3328 monthly sunspot values from predict_sunspots.csv
Window=12, Horizon=6, Train=2650, Test=666
Built: 51 clusters, 32.0x compression, order=2
Forecast quality (661 test windows):
```

step	avg-err	max-err
t+1	17.58	88.10
t+2	19.48	115.00
t+3	21.01	107.10
t+4	21.76	114.50
t+5	22.84	118.40
t+6	24.26	153.20

```
avg confidence: 1.000
```

Data Source Monthly mean total sunspot number from SILSO (World Data Center, Royal Observatory of Belgium). Semicolon-delimited: year, month, decimal_year, sunspot_number, std_dev, obs_count, marker. "Source: WDC-SILSO, Royal Observatory of Belgium, Brussels"

predict_earthquakes

Predict earthquake magnitude from geographic location and depth. This is a spatial prediction problem (not timeseries): given where an earthquake occurs, what magnitude is expected based on historical patterns at nearby locations?

```
./predict_earthquakes.x <usgs_csv> [train_fraction]
```

```
usgs_csv          USGS earthquake catalog CSV (comma-delimited).
train_fraction    Fraction of data for training (default: 0.8).
```

```
./predict_earthquakes.x predict_earthquakes.csv
```

```
Loaded 19619 earthquake events from predict_earthquakes.csv
Inputs: latitude, longitude, depth -> Output: magnitude
Train=15695, Test=3924
Built: 125 clusters, 83.9x compression, order=2
Prediction quality (3924 test events):
```

avg magnitude error:	0.272
max magnitude error:	2.700
avg confidence:	0.649

Data Source USGS Earthquake Hazards Program (public domain, US Government). Comma-delimited: time, latitude, longitude, depth, mag, magType, ...

predict_discharge

River discharge (streamflow) forecasting using sliding-window kNN with day-of-year as an additional input dimension to capture seasonality. Predicts the next 7 days from the previous 14 days.

```
./predict_discharge.x <discharge_tsv> [train_fraction]
```

```
discharge_tsv    USGS NWIS daily discharge (tab-delimited RDB format).
train_fraction   Fraction of data for training (default: 0.8).
```

```
./predict_discharge.x predict_discharge.tsv
```

```
Loaded 9135 daily discharge values from predict_discharge.tsv
```

```
Window=14 (+day-of-year), Horizon=7, Train=7294, Test=1827
```

```
Built: 85 clusters, 58.9x compression, order=2
```

```
Forecast quality (1821 test windows):
```

step	avg-err	max-err
t+1	763.6	22600.0
t+2	895.3	28800.0
t+3	1015.9	29800.0
t+4	1091.7	27100.0
t+5	1162.7	27200.0
t+6	1249.7	27100.0
t+7	1353.2	27200.0
avg confidence:	1.000	

Data Source USGS National Water Information System (public domain, US Government). Colorado River at Lees Ferry, site 09380000. Tab-delimited RDB, comment lines start with #, data columns: agency_cd, site_no, datetime, discharge_value, qualification_code.

How It Works

All samples share the same prediction library (libxs_predict):

1. **predict_params**: Each CSV row is an independent (inputs, outputs) pair. The model learns spatial relationships in the input space and predicts outputs for unseen input combinations.
2. **predict_sunspots**: Each sliding window of W consecutive values becomes an input vector; the next H values become outputs. The model finds historically similar windows and predicts the continuation. The kNN confidence reflects how well the current pattern matches training history.
3. **predict_earthquakes**: Each earthquake event provides (lat, lon, depth) as inputs and magnitude as output. The model finds geographically similar past events and predicts expected magnitude for new locations.
4. **predict_discharge**: Combines temporal sliding-window (14 days) with day-of-year seasonality as an extra input dimension. The model captures both short-term flow dynamics and annual patterns.

The fingerprint automatically determines per-output whether polynomial interpolation or distance-weighted kNN voting is more appropriate. Per-output confidence scores enable the caller to gate predictions and fall back to safe defaults when the model is uncertain.

Registry Microbenchmark

Benchmarks the performance of the LIBXS registry (key-value store) dispatch path under different access patterns and concurrency levels.

Programs

```
./registry.x [total [nrepeat [nthreads]]]
```

Argument	Default	Description
total	10000	Number of unique keys to register (min 2)
nrepeat	10	Repeat iterations for lookup phases
nthreads	max available	Number of OpenMP threads

Measurements:

- Duration to register (insert) all keys into the registry.
- Cold lookup with shuffled access pattern (defeats thread-local cache).
- Cached lookup with sequential/repeating pattern (hits thread-local cache).
- Multi-threaded parallel reads across all threads.
- Contended parallel writes (each thread registers its own key range).
- Mixed read/write: one writer thread while remaining threads read.

The multi-threaded benchmarks require at least 2 threads and are skipped when running single-threaded.

```
./registryf.x
```

Fortran variant with hardcoded parameters (10000 keys, 10 repeats). Measures registration, cold lookup, and cached lookup.

Scaling Behavior

Read-only accesses stay roughly constant in per-op duration due to the thread-local cache. Write accesses are serialized and duration scales with the number of threads.

Rosetta

Demonstrates hierarchical type discovery on opaque binary data. A table of harmonic-series values is encoded as a flat byte blob with no metadata. The analysis rediscovers the element type, record stride, and per-field structure -- recovering mathematical content from anonymous bytes.

What It Shows

The sample proceeds through the levels described in the algebra paper (Section "Hierarchical Type Discovery"):

Step	Tool used	What it finds
Flat probe	libxs_fprint (UNKNOWN)	inconclusive (expected)
Stride sweep	libxs_fprint per-column	f64, stride=6
Shuffle test	libxs_shuffle + fprint	order matters (3-4x)
Field analysis	libxs_fprint per-field	decay rates per column
Sort test	libxs_sort_smooth (GREEDY)	already optimally ordered
Verification	libxs_setdiff_min	0 unmatched, tol=0

Key observations from the output:

- The flat 1-D probe fails because interleaved fields of different scales ($1/n$ mixed with n mixed with $\ln(n)$) look like noise when read sequentially. This motivates the stride sweep.
- The stride sweep requires ALL columns at a candidate width to have decay < 1 before accepting it, which eliminates false positives from partial correlations at wrong strides.
- Shuffle stability confirms that the record order carries genuine structure (decay increases $\sim 4x$ after coprime permutation).
- Per-field analysis reveals:
 - [0] decay=0 -- perfect ramp (sequential index)
 - [1] decay ~ 0.63 -- $1/n$ (decaying but not ultra-smooth)
 - [2] decay ~ 0.20 -- H_n (partial sums, very smooth)
 - [5] decay ~ 0.29 -- converges to Euler-Mascheroni gamma
- GREEDY sort confirms the data is already optimally ordered (the natural 1..64 sequence is the smoothest permutation).

The Data

For $n = 1, 2, \dots, 64$ the table stores six f64 fields per record:

```
[0] n           integer index
[1] 1/n         harmonic term
[2] H_n         partial sum of the harmonic series
[3] ln(n)       natural logarithm
[4] H_n - ln(n) difference (converges to gamma from above)
[5] H_n - ln(n) - 1/(2n)  better gamma estimate
```

Total: 64 records x 6 fields = 384 doubles = 3072 bytes.

Build

```
cd samples/rosetta
make GNU=1
```

Usage

```
./rosetta.x
```

No arguments. The sample is self-contained: it generates the data, encodes it as a flat blob, runs the full analysis, and prints the results.

Example Output

```
ROSETTA: Recovering structure from opaque bytes
Ground truth (hidden from the analysis):
  64 records x 6 fields = 384 doubles (3072 bytes)
  Data: harmonic series H_n and derived quantities.
  Field [5] converges to Euler-Mascheroni gamma.
Level 0: Flat 1-D probe (all bytes as one sequence)
  Input: 3072 opaque bytes
  No type has decay < 1 -- flat stream is not smooth.
  This is expected: interleaved fields of different
  scales look like noise when read sequentially.
Stride sweep: discovering record layout
  Best type:      f64
  Best stride: 6 elements (48 bytes per record)
  Records:       64
  Avg decay:     0.288244
Shuffle stability (record-level)
  Avg decay (original): 0.288244
  Avg decay (shuffled): 1.103839
  Ratio:         3.8x
  -> Record order carries structure.
Per-field decay analysis
[0] n          decay=0.000000
[1] 1/n        decay=0.631512
[2] H_n        decay=0.203226
[3] ln(n)      decay=0.259621
[4] H_n-ln(n)  decay=0.342853
[5] gamma_est  decay=0.292251 (last=0.5771953203, gamma=0.5772156649)
Smoothest: [0] n (perfect ramp, decay=0)
-> This reveals a sequential index column.
Most interesting: [5] gamma_est -- converges to a constant.
GREEDY sort test (64 rows x 6 cols)
  Data is already optimally ordered for row smoothness.
Verification: setdiff(original, blob)
  Unmatched: 0, tolerance: 0.00e+00
  -> Byte-perfect recovery.
```

Why This Is Interesting

From 3072 anonymous bytes the framework discovers:

1. The element type (f64) -- not i32, not f32, not i64.
2. The record structure (6-field records of 48 bytes each).
3. A sequential index column (field [0], decay = 0).
4. A column converging to Euler-Mascheroni gamma (field [5]).
5. That the natural ordering is already optimal (no resorting).

No metadata, no format knowledge, no human guidance. The hierarchical composition -- stride sweep with all-columns-must-pass filtering, shuffle stability, per-field decay ranking -- is what makes this possible. Any single tool alone would either fail (flat probe) or produce ambiguous results (stride sweep without the all-columns requirement accepts wrong strides).

Memory Allocation (Microbenchmark)

Benchmarks pooled scratch memory allocation (`libxs_malloc` / `libxs_free`) against the standard C library `malloc` / `free`. The pool-based allocator recycles memory after an initial warm-up, avoiding repeated system calls in steady state. A Fortran variant compares `libxs_malloc` against Fortran `ALLOCATE` / `DEALLOCATE`.

Building

```
cd samples/scratch
make GNU=1
```

Produces `scratch.x` (C) and, if a Fortran compiler is found, `scratchf.x`.

Usage (C)

```
./scratch.x [ncycles [max_nactive [nthreads]]]
```

Argument	Default	Description
ncycles	100	Number of allocation/deallocation cycles
max_nactive	4	Max concurrent live allocations per cycle (max 24)
nthreads	1	OpenMP threads (clamped to <code>omp_get_max_threads</code>)

Environment Variables

Variable	Default	Description
CHECK	0	Non-zero: <code>memset</code> each allocation (validates writeable)

Compile-Time Knobs

Macro	Default	Description
<code>MAX_MALLOC_MB</code>	100	Upper bound on individual allocation size in MB
<code>MAX_MALLOC_N</code>	24	Array size for random-number table and alloc slots

```
## default: 100 cycles, up to 4 active allocations, 1 thread
./scratch.x
```

```
## heavy load: 500 cycles, up to 12 active allocations, 4 threads
./scratch.x 500 12 4
```

```
## validate pages are writable
CHECK=1 ./scratch.x 43 8 4
```

Usage (Fortran)

```
./scratchf.x [mbytes [nrepeat]]
```

Argument	Default	Description
mbytes	100	Allocation size in MB
nrepeat	20	Number of repetitions

Single-threaded benchmark comparing `libxs_malloc` / `libxs_free` against Fortran `ALLOCATE` / `DEALLOCATE`. Reports per-iteration times (ms) and an average speedup ratio.

What Is Measured

Each cycle draws a random number of allocations (1 to `max_nactive`) with random sizes (1 to `MAX_MALLOC_MB` MB each). The cycle allocates all buffers, optionally touches them (`CHECK`), then frees them.

Both paths use the same randomized sequence. An untimed warm-up cycle runs before the measured loops. Reported metrics:

- calls/s (kHz) -- allocation+free throughput for each allocator
- Scratch size -- high-water mark of the pool (MB)
- Malloc size -- peak aggregate allocation per cycle (MB)
- Scratch Speedup -- ratio of pool throughput to `stdlib` throughput
- Fair -- size-adjusted speedup: $(\text{malloc_size} / \text{scratch_size}) * \text{speedup}$

Shuffle

Benchmarks three shuffling strategies and compares their throughput:

Label	Method
RNG-shuffle	Fisher-Yates via <code>libxs_rng_u32</code> (reference baseline)
DS1-shuffle	<code>libxs_shuffle</code> -- deterministic in-place (coprime stride)
DS2-shuffle	<code>libxs_shuffle2</code> -- deterministic out-of-place (src to dst)

Each iteration works on an array of unsigned integers initialized to 0..n-1. On the final iteration the shuffled result can optionally be written as an MHD image file or analyzed for randomness quality.

Build

```
cd samples/shuffle
make GNU=1
```

Usage

```
./shuffle.x [nelems [elemsize [niters [repeat]]]]
```

Positional	Default	Description
nelems	64 MB / elemsize	Number of elements
elemsize	4	Element size in bytes (1, 2, 4, or 8)
niters	1	Shuffle passes per timed measurement
repeat	3	Timed iterations (first is warmup)

Environment Variables

Variable	Default	Description
RANDOM	0	Non-zero: count inversions and report randomness (rand=%)
SPLIT	1	Partitioning depth for the STATS metrics
STATS	0	Non-zero: print partition imbalance (imb) and distance (dst)

```
## Default run (64 MB of 4-byte elements, 3 repeats)
./shuffle.x

## 1M elements of 8 bytes, 4 shuffle passes, 5 repeats
./shuffle.x 1000000 8 4 5

## Enable randomness quality metric
RANDOM=1 ./shuffle.x 10000 4 1 3

## Enable partition statistics with split depth 2
STATS=1 SPLIT=2 ./shuffle.x
```

What Is Measured

Reported bandwidth is 2 * data-size / time (in MB/s). The factor of two accounts for reading and writing each element during the shuffle. The first iteration is excluded from the average.

Optional quality metrics:

- rand -- Enabled by RANDOM=1. Inversions are counted via merge-sort and compared against the expected count for a random permutation ($n*(n-1)/4$) to give a percentage.
- dst -- Manhattan distance of the element sum from the expected uniform value, split into hierarchical partitions (SPLIT).
- imb -- Partition imbalance, measuring how unevenly element sums are distributed across sub-partitions (SPLIT).

MHD Image Output When RANDOM is not set and the element size is 1, 2, 4, or 8 bytes, the final shuffled array is written as an MHD image per method (shuffle_rng.mhd, shuffle_ds1.mhd, shuffle_ds2.mhd). The images are shaped close to square ($\text{isqrt}(n) \times n/\text{isqrt}(n)$) and converted to unsigned 8-bit via modulus.

Compile-Time Knobs

Variable	Default	Description
OMP	0	Enable OpenMP (not used by this sample)
SYM	1	Include debug symbols (-g)
BUBBLE_SORT	undefined	Define (-DBUBBLE_SORT) for O(n^2) inversion counter

Synchronization Primitives

Micro-benchmark for the lock implementations provided by LIBXS (`libxs_sync.h`). Measures single-thread latency (uncontended acquire/release) and multi-thread throughput (mixed read/write workload) for every compiled lock kind:

Lock kind	Description
LIBXS_LOCK_DEFAULT	Compile-time default (typically atomic)
LIBXS_LOCK_SPINLOCK	OS-native or CAS-based spin lock (if available)
LIBXS_LOCK_MUTEX	OS-native mutex / <code>pthread_mutex_t</code> (if available)
LIBXS_LOCK_RWLOCK	Reader/writer lock / <code>pthread_rwlock_t</code>

The default lock is always benchmarked. The remaining kinds are conditionally compiled depending on platform support.

Build

```
cd samples/sync
make GNU=1
```

OpenMP is enabled by default (`OMP=1`) for multi-threaded tests.

Run

```
./sync.x [nthreads] [wratio%] [work_r] [work_w] [nlat] [ntpt]
```

Argument	Default	Description
nthreads	all available	Number of OpenMP threads
wratio%	5	Percentage of write operations (0-100)
work_r	100	Simulated work inside read-critical section (cy)
work_w	10 * work_r	Simulated work inside write-critical section
nlat	2000000	Iterations for latency measurement
ntpt	10000	Iterations per thread for throughput measurement

```
./sync.x 4 5 100 1000
```

```
LIBXS: default lock-kind "atomic" (Other)
```

```
Latency and throughput of "atomic" (default) for nthreads=4 wratio=5% ...
ro-latency: 11 ns (call/s 91 MHz, 33 cycles)
rw-latency: 11 ns (call/s 90 MHz, 33 cycles)
throughput: 0 us (call/s 9128 kHz, 328 cycles)
```

Measurement Details

- RO-latency: uncontended read-lock acquire/release pairs (4x unrolled), reported as nanoseconds per operation and TSC cycles.
- RW-latency: uncontended write-lock acquire/release pairs (4x unrolled).
- Throughput: all threads run a mixed read/write workload governed by wratio%. Simulated work inside the critical section is subtracted so only synchronization overhead is reported.

SYRK / SYR2K Sample

Demonstrates symmetric rank-k and rank-2k updates using the LIBXS dispatch-and-call model. The sample validates correctness against a plain Fortran reference and reports performance.

Operations

SYRK: $C := \alpha * A * A^T + \beta * C$ (lower triangle) SYR2K: $C := \alpha * (A * B^T + B * A^T) + \beta * C$ (upper triangle)

Only the specified triangle of C is written; the other triangle is left untouched.

Build

```
make
```

Requires a Fortran compiler (gfortran, ifort, or ifx). The Makefile picks up the compiler from the top-level Makefile.inc. MKL or BLAS linkage is enabled (BLAS=1) so that dispatch can use JIT-compiled kernels when available.

Run

```
./syrkf.x [N [K [nrepeat]]]
```

Arguments (all optional, positional):

N	Matrix dimension of C (N x N).	Default: 64
K	Inner dimension (columns of A).	Default: N
nrepeat	Number of timed repetitions.	Default: 100

Example Output

```
syrk(F): N=64 K=64 nrepeat=100

--- libxs_syrk (lower) ---
max error (lower): 0.00000E+00

--- libxs_syr2k (upper) ---
max error (upper): 0.00000E+00

--- SYRK performance ---
time:      0.002 s (100 calls)
perf:      28.4 GFLOPS/s
```

Notes

- The dispatch step (`libxs_syrk_dispatch` / `libxs_syr2k_dispatch`) returns a pointer to a registry-owned config. This pointer remains valid until `libxs_finalize` or the registry is destroyed. There is no need to release it manually.
- Internally, SYRK/SYR2K decompose into GEMM tiles on the diagonal and off-diagonal blocks. The dispatched GEMM kernel (MKL JIT, LIBXSMM, or fallback BLAS) handles the inner loop.
- Scratch memory for the temporary full-panel product is managed via a thread-local buffer that grows on demand and is freed at finalization.